

Can Supportive Coaching Behaviors Facilitate Boundary Spanning and Raise Job Satisfaction? An Indirect-Effects Model

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As organizations encounter unpredictable external environments, expectations are changing for managers and team leaders toward providing more facilitative, less directive coaching in order to stimulate more flexibility and adaptability. Prior research has underscored the role of team leader supportive coaching behaviors in reinforcing and growing team member capabilities to work independently from the leader. What is not yet understood, however, is if and how supportive coaching behaviors relate to team member engagement in boundary-spanning behaviors, which are team member efforts to establish and maintain relationships with key parties external to their team. This study examines how team leaders' supportive coaching of boundary spanning—defined as a set of behaviors that (a) encourages team members to engage in frequent and open collaborations with key parties external to the team and (b) grants team members the latitude to engage external parties in ways the team members deem necessary—relates to team member boundary-spanning behavior and job satisfaction through a motivational pathway. We test a hypothesized indirect-effects model within a field sample of 256 engineering employees working in teams. Results indicate that leaders' supportive coaching of boundary spanning is positively related to team member boundary-spanning self-efficacy, which in turn is positively related to team member engagement in boundary-spanning behavior for their teams. Also, team members who reported more boundary-spanning behavior experienced greater job satisfaction. The findings have important implications for organizations, team leaders, and team members.

Keywords: supportive coaching; boundary-spanning behavior; job satisfaction; mediation

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Rapid, constant, and disruptive change is now the norm [in organizations]. . . . Twenty-first century managers simply don't (and can't!) have all the right answers. . . . The role of the manager, in short, is becoming that of a coach. (Ibarra & Scoular, 2019: 112)

Ibarra and Scoular's (2019) research captures a shift in organizations: Leaders are expected to provide more facilitative, less directive coaching to employees, who are engaged in increasingly dynamic and complex work environments. Reflective of this trend, some team leaders engage in "supportive coaching" behaviors that encourage and empower team members to work independently from the leader and to grow their capacity to self-manage the team processes that drive team and organizational performance (see Carson, Tesluk, & Marrone, 2007; Morgeson, 2005; Wageman, 2001). One team member activity that is critical for modern organizations is *boundary spanning*. Boundary-spanning behaviors are team member efforts to establish and maintain relationships with key entities external to their working unit for the purposes of achieving work-unit tasks and objectives (Ancona & Caldwell, 1992). When a team member solicits resources from upper-level management, he or she is engaged in boundary-spanning behavior. Other examples of boundary-spanning behavior include negotiating deliverable deadlines directly with another working unit and acquiring needed technical expertise from external subject-matter experts. These efforts can yield markedly higher client satisfaction (Marrone, Tesluk, & Carson, 2007), facilitate team learning (Edmondson, 2003), enhance organizational innovation (Hargadon, 1998), and increase sales margins (Gladstein, 1984).

Given the importance of boundary spanning, it is surprising that we do not yet understand *how* team leaders can support and motivate their members to engage in these behaviors. We do not know how (or if) a team leader's choice to "act as coach"—to be more facilitative and less directive—relates positively to team member boundary spanning. Addressing this gap is significant because enacting team boundary spanning has shifted away from team leaders and toward the team itself. Team members are increasingly expected to represent team goals and progress directly to stakeholders, incorporate diverse expertise from other functional areas, and coordinate their interdependencies with other work groups (see Ancona & Bresman, 2007; Cross et al., 2000; Marrone, 2010; Yan & Louis, 1999). Understanding the leader's efforts to support team member engagement in these activities is thus ever more important (Tesluk & Mathieu, 1999). Also, boundary spanning is notably challenging for team members to execute and may require specific leadership support. For example, members who boundary span report role overload and burnout due to conflicting expectations and demands placed on them by diverse external parties, which may relate to poor work experiences and lower job attitudes (Marrone et al., 2007; Singh, Goolsby, & Rhoads, 1994). Leaders might be able to help team members overcome various deterrents to their engagement in this behavior, however, by emphasizing cross-boundary engagement and encouraging boundary-spanning behavior.

In the current study, we therefore examine the relationship of team leader coaching behaviors on team member engagement in boundary spanning and team member job satisfaction. We focus specifically on team leaders' supportive coaching of boundary spanning, which we conceptualize as a set of behaviors that (a) encourages team members to engage in frequent and open collaborations with key parties external to the team and (b) grants team members

the latitude to engage external parties in ways that the team members deem are necessary. We use social cognitive theory (e.g., Bandura, 1986, 1997, 2012) and Hackman and Wageman's (2005) theory of team coaching as the foundation of our arguments, and we examine whether boundary-spanning self-efficacy (Marrone et al., 2007) serves as an important motivational mechanism through which supportive-coaching behaviors relate to team member boundary spanning. We hypothesize that team leader supportive-coaching behaviors relate to higher member beliefs in their boundary-spanning capabilities, these beliefs relate to members' engagement in boundary-spanning behaviors, and these behaviors relate to higher levels of job satisfaction.

This study makes several important contributions. First, we clarify the relationship between supportive-coaching behavior and team member boundary-spanning behavior, which is nearly absent in the work teams coaching literature in the organizational behavior domain. Extant studies have focused on the impacts of team leader supportive-coaching behaviors on internal team dynamics, yet increasing demands are placed on team members to establish and maintain relationships *across* team boundaries. Second, our findings offer an important theoretical extension regarding the effective leadership of externally dependent work units by extending foundational work in the boundary-spanning literature that previously highlighted directive team leader behaviors. We explore the role of supportive-coaching behaviors because boundary spanning is notably challenging for team members to execute and members may seek out and benefit from specific leadership support in this domain. Third, this study advances understanding of how supportive-coaching behavior positively relates to team members' boundary-spanning behavior. Specifying and testing this underlying motivational mechanism answers recent calls for greater theoretical understanding of the antecedents to team member boundary-spanning behavior (Joshi, Pandey, & Han, 2009). Fourth, we examine the relationship between boundary-spanning behavior and job satisfaction in an attempt to elucidate prior boundary-spanning research, where classic arguments for a positive association between boundary-spanning behavior and job satisfaction (e.g., Aldrich & Herker, 1977; Hallenbeck, Hautaluoma, & Bates, 1999; Keller, 1978) have been called into question by some more recent empirical findings (e.g., Marrone et al., 2007; Ramarajan, Bezrukova, Jehn, & Euwema, 2011). Last, this study advances understanding of the ongoing supportive-coaching behaviors exhibited by organizational managers and team leaders. In so doing, we contribute to calls for deeper exploration of managerial coaching dimensions, mechanisms, and effects (e.g., Dahling, Taylor, Chau, & Dwight, 2016).

Theoretical Background

Team Member Boundary-Spanning Behavior

As noted already, boundary-spanning behaviors are defined as the efforts made by employees to establish and maintain relationships with key entities external to one's own working unit for the purposes of achieving work unit tasks and objectives (Ancona & Caldwell, 1992; Marrone et al., 2007). The overarching function of these behaviors is to manage the work unit's relationships with its external environment (Ancona, 1990; Leifer & Delbecq, 1978). In considering work teams as subsystems embedded in their larger organizational system, the necessity for work teams to engage externally becomes readily apparent. Work teams depend upon their organizational environment for "human, technological, and financial information,

for information and power, and eventually for [their] very existence" (Friedlander, 1987: 303). In turn, the operations and output of the team further constitutes the system (Friedlander, 1987). Beyond the organizational environment, work teams and their members also contribute to, and extract resources from, clients, suppliers, and other organizations residing across organizational boundaries (Rosenkopf & Nerkar, 2001; Takeishi, 2001). The active management of a working unit's relationships to key parties in its environment enables the team to secure resources, exert power, obtain information (Aldrich & Herker, 1977), and experiment and collaborate with other work units (Ancona, 1990). As such, member boundary spanning underlies the survival and success of the team and its organization.

Foundational empirical research in the work teams domain by Ancona and Caldwell (1988, 1992) identified three primary types of boundary-spanning behaviors that are executed by work-unit members on the team's behalf. These behaviors aim to (a) represent one's work unit in a positive way to external entities in order to gain further resources and/or protect the unit from undue and adverse pressures (referred to as *ambassadorial* behaviors), (b) scan actively for resources and expertise embedded in external entities (referred to as *scanning* activities), and (c) manage interdependencies across work teams, such as informing others of the work unit's task progress (referred to as *task coordination* activities). Later research supports that engaging in these behaviors significantly affects the performance of the focal working unit, the organization, and other interdependent entities (for related discussions of boundary-spanning behaviors and outcomes, see Cross et al., 2000; Faraj & Yan, 2009; Marrone, 2010; Marrone et al., 2018; Somech & Khalaili, 2014). For example, ambassadorial behaviors can elevate the unit's ability to secure needed resources to achieve organizational objectives (Ancona & Caldwell, 1992; Benoliel & Somech, 2015; Druskat & Wheeler, 2004), and scanning behaviors can result in greater organizational learning and innovation (Lorenzoni & Lipparini, 1999; Mors, 2010).

The benefits of boundary-spanning behavior notwithstanding, significant variance exists in the extent to which team members will engage in these activities. Early in-depth qualitative studies conducted by Ancona (1990) and Ancona and Caldwell (1992) demonstrated that within some teams, members exhibit high levels of boundary-spanning activity, whereas in other teams, members exhibit notably low levels and remain comparatively more isolated from external parties relevant to the team's task work. Research has not yet determined which antecedent variables are most predictive of this variance, though these qualitative depictions illuminated leadership as an important factor to consider (see also Marrone, 2010). Team leaders who explicitly communicated strategies to their teams that emphasized early external team activity were uniquely successful in influencing team members to engage in important boundary-spanning behaviors that resulted in the team's higher effectiveness. These team leaders understood the need for boundary spanning, placed priority on this activity, and directed members' behaviors accordingly.

In contrast, when team leaders did not prioritize and direct the boundary-spanning function, team members demonstrated either passive reactivity to the team's external environment or unilaterally imposed their team's ideas on external parties without an active consideration of the roles, contributions, or objectives of the team's existing and potential contextual relationships (Ancona, 1990). These teams were significantly less effective overall. Empirical research conducted by Hackman (2002) and colleagues likewise concluded that without team leader explicit direction to interact externally, team members tended to

exhibit passive reactivity to their environment. It may be that without team leader direction and encouragement, the demands associated with interacting beyond the team's boundary will reduce the willingness of the team's members to engage in boundary-spanning behaviors. Indeed, boundary spanners have reported high levels of role overload as a result of continued exposure to diverse parties' perspectives and objectives (Marrone et al., 2007) and negative job experiences (Ramarajan et al., 2011).

Supportive Coaching of Boundary Spanning

An alternative approach for team leaders who wish to encourage—rather than direct or ignore—boundary spanning may be the use of supportive-coaching behaviors. Supportive-coaching behaviors are team leader efforts to reinforce the team's competencies and capabilities independently from the leader (Morgeson, 2005). It is important to note that we follow the teams literature within organizational behavior (e.g., Carson et al., 2007; Morgeson, 2005; Wageman, 2001) in examining the behaviors of “team leads,” or first-line team managers. These managers are internal to the organization and have supervisory authority and leadership responsibility with respect to the teams. This approach is aligned with broader managerial coaching studies, which increasingly emphasize the ongoing roles managers play in supporting the development and growth of employees (e.g., Dahling et al., 2016; Ellinger, 2013; Hui & Sue-Chan, 2018). A primary aspect of supportive-coaching behaviors is the leader's empowerment of the team's members (Arnold, Arad, Rhoades, & Drasgow, 2000; Burke, Stagl, Klein, Goodwin, Salas, & Halpin, 2006; Weer, DiRenzo, & Shipper, 2016). For example, Carson et al. (2007) found that leaders who supported their teams with coaching behaviors established internal team conditions that enhanced team members' voice, empowerment, and shared leadership. Similarly, Farh and Chen (2018) found that coaching promoted employee voice in a field study of surgical teams. Supportive coaching empowers team members through encouraging team self-management and providing team members with latitude and authority so that they may act independently and free from leader interference (Farh & Chen, 2018; Morgeson, 2005; Wageman, 2001). As Morgeson (2005: 502) noted, “Practically, this would involve providing reinforcement to the team but not actually becoming involved in the task performance itself.”

Supportive coaching of boundary spanning thus reinforces the team's capabilities to carry out boundary-spanning behaviors without direct intervention from their leader. We conceptualize supportive coaching of boundary spanning as a set of behaviors that (a) encourages team members to engage in frequent and open collaborations with key parties external to the team and (b) grants team members the latitude to engage external parties in ways that the team members deem are necessary. The combination of encouragement and responsibility is vital to the supportive-coaching approach as it both guides team members toward engagement in productive boundary-spanning tasks and empowers team members with the authority and control to carry out these actions independently from their leader. Empowered team members, in turn, realize increased self-efficacy (Arnold et al., 2000), satisfaction (Ellinger, Ellinger, & Keller, 2003; Morgeson, 2005), and learning (Burke et al., 2006; Edmondson, 2003) that, in turn, stimulate greater team performance. Indeed, numerous empirical findings suggest that team effectiveness is positively associated with supportive coaching behaviors (see Buljac-Samardzic & van Woerkom, 2015; Burke et al., 2006; Liu, Pirola-Merlo, Yang,

& Huang, 2009; Rousseau, Aube', & Tremblay, 2013; Weer et al., 2016), and preliminary evidence suggests that supportive-coaching behaviors may positively influence teams to actively coordinate with other work groups (Edmondson, 2003; Tesluk & Mathieu, 1999).

Supportive-coaching behaviors influence work-unit effectiveness through motivational means (see Burke et al., 2006) that enhance the effort that team members will exert toward work tasks. Hackman and Wageman's (2005) theory of team coaching specifies the motivational role that team leader coaching behaviors plays in supporting team effectiveness. The authors suggest that when coaching efforts reinforce team capabilities and encourage self-management, as is the intentional aim of supportive coaching, team leaders communicate optimism regarding the team's capabilities that motivates team members to contribute greater efforts toward their collective team-related tasks. In support of this theory, Liu et al.'s (2009) study of 137 research and development teams in Taiwan demonstrated that increases in team member efforts significantly mediated the positive influence of team coaching on team effectiveness. Additionally, Weer et al. (2016) studied 714 managers and their teams over 54 months and found a positive and significant link between supportive coaching and the strength of team members' effort toward team tasks (labeled as team commitment). Rousseau et al. (2013) also found that supportive coaching motivated greater team commitment, which in turn enhanced team innovation, for teams in a public safety organization.

Hypothesis Development

In the sections that follow, we contend that the motivating influence of supportive coaching is most likely to occur through a salient mediating mechanism—boundary-spanning self-efficacy (i.e., team members' views about their ability to perform boundary-spanning tasks effectively; Marrone et al., 2007)—that enhances team members' views about their boundary-spanning capabilities. We hypothesize that supportive coaching of boundary spanning indirectly relates to team member job satisfaction through higher boundary-spanning self-efficacy and greater engagement in boundary-spanning behaviors.

Supportive Coaching of Boundary Spanning and Team Member Boundary-Spanning Self-Efficacy

A team leader's supportive-coaching behaviors likely facilitate greater engagement in member boundary-spanning behavior through enhancing team member perceptions of their boundary-spanning self-efficacy. Classic work in social cognitive theory (e.g., Bandura, 1986, 1997) emphasizes the critical role of self-efficacy (defined as the belief in one's capability to organize and execute the course of action required to succeed; Bandura, 1997; Hardy, Day, & Steele, 2019) as a factor in the self-regulation of motivation and behavior. Numerous studies over the past several decades find that self-efficacy has positive effects on work performance (e.g., Stajkovic & Luthans, 1998). Importantly, self-efficacy perceptions likely "vary across activity domains and situational conditions rather than manifest uniformly . . . in the likeness of a general trait" (Bandura, 2012:13). In keeping with prior research that considers boundary-spanning behavior as an activity domain (e.g., Lanaj & Hollenbeck, 2015; Marrone et al., 2007), we examine the role of supportive-coaching behaviors in building boundary-spanning self-efficacy, which we in turn expect will result in higher levels of

boundary-spanning behavior. As noted earlier, we define boundary-spanning self-efficacy as a team member's belief in his or her ability to "successfully establish and manage relationships with important parties external to his or her team" (Marrone et al., 2007: 1426).

Following Bandura's (1997, 2012) identification of verbal persuasion as one primary source of self-efficacy, team members will likely internalize the verbal support and responsibility regarding boundary spanning given to them by authority figures, thereby making it easier to hold and sustain affirming self-beliefs about this action. Indeed, Boyer, Zollo, Thompson, Vancouver, Shewring, and Sims (2000) found in a meta-analysis that verbal persuasion is an especially impactful antecedent of self-efficacy, as it can raise (or lower) self-efficacy beliefs independent of performance. Individuals' expectations of themselves are highly influenced by the expectations of their leaders (Mitchell & Daniels, 2003) and particularly so because these expectations are a salient aspect of the individual's immediate social environment from which sensemaking occurs (Weick, 1995). Furthermore, team members are additionally motivated to internalize the positive expectations signaled by their leader's supportive coaching to maintain consistent views about themselves that are positive and coherent with signals from their external environment (Salancik & Pfeffer, 1978).

Though studies of team leader coaching behaviors have not yet addressed boundary-spanning self-efficacy, scholars have theorized that self-efficacy in other task domains is a primary mechanism through which supportive coaches exert motivational influence (e.g., Dahling et al., 2016; Manz & Sims, 1987). As one example, Pousa and Mathieu (2015) found that managerial coaching significantly enhanced employee self-efficacy, which then fully mediated coaching's effect on changes in employee behavioral performance (i.e., actions employees took to achieve desirable performance results) in a sample of 122 financial advisors. We hypothesize the following:

Hypothesis 1: Supportive coaching of boundary spanning is positively related to team member boundary-spanning self-efficacy.

Team Member Boundary-Spanning Self-Efficacy and Boundary-Spanning Behavior

Team members with higher levels of efficacy regarding their ability to boundary span may be more likely to undertake subsequent boundary-spanning efforts and may persist longer in their efforts despite any difficulties that may arise. This occurs through a number of self-regulatory pathways associated with more positive self-efficacy beliefs (Bouffard-Bouchard, 1990). Bandura (2012: 13) noted that self-efficacy beliefs contribute to choice processes as individuals decide on their course of action because perceptions of self-efficacy "affect the slate of options people consider and the choices they make at important decision points." Self-enabling beliefs allow team members to take greater risks in engaging in boundary-spanning activity because efficacious team members view challenges as opportunities to learn and excel rather than as threats or insurmountable obstacles (Bandura, 1997, 2012). Furthermore, those with high levels of self-efficacy interpret their bodily reactions to stressful work tasks adaptively (Bandura, 1997, 2012); team members with higher boundary-spanning self-efficacy may experience less anxiety associated with boundary-spanning behavior than those with lower levels of boundary-spanning self-efficacy (Bandura, 1997). These

gains in stress resistance should allow for greater persistence in boundary-spanning activity over time (Hobfoll, 1989). As team members are encouraged by their leaders and persuaded verbally of their boundary-spanning capabilities, they are likely to exert greater effort toward boundary-spanning activities and sustain those efforts over time, thereby affirming their beliefs that such behaviors are valued, supported, and likely to be effective.

In the boundary-spanning domain, one existing empirical study supports the link between efficacy and behavior. Boundary-spanning self-efficacy emerged as a strong motivational driver of team member boundary-spanning behavior in a study of MBA consulting teams (Marrone et al., 2007). The authors surmised that boundary-spanning self-efficacy led to higher goal setting and more realistic task performance strategies, which promoted both boundary-spanning activity and success. Importantly, Marrone et al. (2007) is the only published study providing evidence of a positive relationship between boundary-spanning self-efficacy and boundary-spanning behavior. Moreover, their work examined this relationship in one team-based context (i.e., consulting team members) using a sample of MBA students. Also, Marrone et al. (2007) did not consider the role of leadership or supportive-coaching behaviors in helping to increase boundary-spanning activity through positively influencing boundary-spanning self-efficacy beliefs; one contribution of the current research is to examine this indirect effect. Based on the aforementioned theoretical arguments, and rooted in Bandura's (1986, 1997, 2012) work and the existing empirical evidence, we hypothesize the following:

Hypothesis 2: Team member boundary-spanning self-efficacy will be positively related to team member boundary-spanning behavior.

Supportive Coaching, Team Member Boundary-Spanning Behavior, and Team Member Job Satisfaction

Prior research is somewhat ambiguous regarding the effects of boundary-spanning behaviors at the individual level. Although boundary-spanning behaviors can potentially contribute to role overload and burnout (e.g., Marrone et al., 2007; Singh et al., 1994), engagement in team member boundary-spanning behavior may also elevate team member job satisfaction. Job satisfaction is "a pleasurable or positive emotional state" that is "a function of the perceived relationship between what one wants from a job and what one perceives it is offering" (Locke, 1976: 1300). To the extent that individuals perceive that their job expectations are fulfilled, they will experience positive job satisfaction (Bacharach, Bamberger, & Conley, 1991). Boundary-spanning behaviors may elevate job satisfaction because engaging in boundary-spanning directly enables team members to achieve productive work-related tasks, such as acquiring needed resources for their team and successfully implementing team ideas (see Ancona, 1990; Ancona & Caldwell, 1992; Friedlander, 1987). Because goal achievement meets individuals' basic needs for autonomy and competence (Deci & Ryan, 2000) and satisfies team needs for performance and survival (Morgeson, DeRue, & Karam, 2010), boundary-spanning behaviors that facilitate goal attainment or other desirable outcomes can enable greater job satisfaction. For instance, Aldrich and Herker (1977) theorized that because boundary-spanning actions enable individuals to gain power and exercise control over their environment in ways that "improve their bargaining

position,” their job satisfaction should increase. Several empirical studies have supported these ideas, demonstrating that contrary to prior theories, boundary-spanning activity was unrelated to role stressors and positively related to job satisfaction (e.g., Keller, 1978; Keller & Holland, 1975). Our arguments are consistent with those of Locke (1970) and others who viewed job satisfaction as a function of performance leading to valued outcomes that are satisfying to individuals (see a review of studies by Judge, Thoresen, Bono, & Patton, 2001). We thus hypothesize a direct effect of team member boundary-spanning behavior on team member job satisfaction:

Hypothesis 3: Team member boundary-spanning behavior will be positively related to team member job satisfaction.

Moreover, leader encouragement of boundary-spanning behaviors may make salient their importance for team members, even if these behaviors do not directly assist team members in goal attainment. More specifically, when leaders explicitly encourage and emphasize engagement in boundary-spanning behaviors, and team members respond with higher levels of these behaviors, leaders may validate team members’ perceived and actual capabilities to meet their job expectations. Ultimately, this process could result in higher team member job satisfaction.

In sum, supportive coaching of boundary spanning signals leader confidence in team members’ capabilities regarding boundary-spanning behaviors that are proximally relevant to desired job outcomes, such as relationship building and coordination of mutually beneficial deadlines across interdependent teams. As such, the resulting improvements in job attitudes (i.e., higher job satisfaction) from this type of supportive leadership are likely to be realized through team members’ higher engagement specifically in those boundary-spanning behaviors. It is important to note, therefore, that supportive-coaching behaviors are not likely to directly or immediately result in a rise in job satisfaction. Rather, we suspect that as efficacious team members explore boundary-spanning tasks independently from their leader, they experience opportunities to observe the consequences of their efforts, are empowered to adapt their boundary-spanning behaviors as appropriate based on what they observe, and learn to respond to unique circumstances with external parties as those issues arise, resulting in an enacted mastery of boundary spanning that is satisfying and productive (for related arguments, see Hui & Sue-Chan, 2018). We thus hypothesize an overall indirect effect of supportive coaching of boundary spanning on job satisfaction through team member boundary-spanning self-efficacy and boundary-spanning behavior:

Hypothesis 4: Supportive coaching of boundary spanning will have an indirect relationship with job satisfaction via team member boundary-spanning self-efficacy and team member boundary-spanning behavior serially.

Method

To test our hypotheses, we collected field data via surveys and used a variety of analytical tools, including Mplus (1998-2017), which allows for an assessment of direct and indirect effects while including multilevel model specifications. The analysis of field data for this study is important in that it allowed us to observe and understand the role that

supportive-coaching behaviors may play on a daily basis within intact working teams in a real organization in encouraging boundary-spanning behavior and, as a result, positively relating to higher job satisfaction in team members. We describe further details of the sample, analytical techniques, and results of preliminary analyses for assessing nonindependence of observations, missing data assumptions, and overall measurement model fit next.

Sample and Procedure

A large division of a manufacturing and engineering firm located in the United States served as the host for the research. Six hundred and eighty-six employees (nonmanagement team members) were contacted via email and asked to participate in a voluntary study involving two waves of data collection separated by a 3-month gap.¹ This time interval ensured minimal disruption for the employees (based on discussions with management) and allowed the research team to manage concerns about survey length and common-method bias. Two hundred and sixty-two team members responded to the first survey (which included measures of supportive coaching and boundary-spanning self-efficacy) for a first-wave response rate of 38.2%. One hundred and seventy-nine of these individuals also responded to the second survey (which included measures of boundary-spanning behavior and job satisfaction) for a response rate to both surveys of 26.1%. Of the 179 who also responded to the Wave 2 survey, 170 included mostly complete data. The average age of this matched sample was 41.92 years ($SD = 11.11$), approximately 90.0% of the respondents were male, 54.3% identified the United States as their country of origin (geographical areas included India/Asia [14.7%], North America [58.2%], Russia [11.2%], and western Europe [12.9%]), and on average, participants had worked at the organization for 9.04 ($SD = 8.75$) years. Approximately 64.7% of the respondents identified as White. Other ethnicities included African American/Black (1.8%), Asian (22.4%), Hispanic/Latino (5.9%), Native American (1.2%), Pacific Islander (1.2%), and Other (5.3%, including Russian and multiethnic). Though we used the matched sample to assess our measures and measurement models, we used all available data to assess structural model relationships.

Following survey data collection, two analyses were conducted to determine possible differences between Wave 1 and Wave 2 survey respondents. First, *t* tests were conducted on demographic characteristics to determine if sampling bias contributed to differences across the two waves of data acquisition. Results indicated no significant differences in age, gender, or education. Second, Goodman and Blum's (1996) approach was followed to determine if sampling characteristics contributed to nonrandom bias. Specifically, we used logistic regression to assess attrition bias because assessing bias on the basis of demographic differences alone is insufficient, and it is critical to examine whether respondents also differ on substantive model variables (Goodman & Blum, 1996). We first created a dependent variable distinguishing those who completed both surveys from those who completed only the first survey. The substantive variables collected in the Wave 1 survey were then regressed onto this dependent measure. Results indicated no nonrandom sampling bias; we thus concluded that subject attrition would not bias subsequent measurement data analyses.

Measures

The particular items and rating scales associated with each measure are in the appendix.

Supportive coaching of boundary spanning (herein labeled as “supportive coaching”). We relied on previous studies assessing supportive-coaching behaviors of encouragement and empowerment (e.g., Morgeson, 2005), as well as our discussions and interviews of organizational personnel, to develop a four-item scale for the purposes of this study that asked respondents to rate the extent to which they agreed that their lead encouraged and empowered employee engagement in boundary spanning. We also followed prior research supporting the use of employee ratings of leader coaching behavior. For instance, Ellinger et al. (2003) found that self-reported ratings of supervisor coaching behaviors were significantly different (higher and inflated) than employee ratings of their supervisor’s supportive-coaching behavior. Leader self-reported assessments of coaching did not relate to employee attitudes, whereas employee ratings were associated with higher job satisfaction and commitment (Thornhill & Saunders, 1998). Furthermore, though not measuring leader coaching, Karasek’s (1979) landmark study of job demands “suggests that self-assessments are a reasonably accurate measure of job decision latitude.” Because we were interested in employee perceptions of supportive coaching, aggregation to the unit level of analysis was not appropriate.

Boundary-spanning self-efficacy. A previously established six-item scale from Marrone et al. (2007) was used to assess the extent to which team members agreed with statements regarding their confidence in their ability to effectively carry out and manage boundary-spanning behaviors. We chose to use this measure for two specific reasons. First, the measure we used has been validated in previous boundary-spanning research that found its reliability to be .92; also, the measure was predictive of boundary-spanning behavior (Marrone et al., 2007). Second, previous research has found that Likert-type and traditional measures of self-efficacy have similar reliability, predictive validity, factor structures, and discriminability and therefore has recommended using Likert-type measures over traditional strength/magnitude measures given their superior practicality (Maurer & Pierce, 1998).

Boundary-spanning behavior. We closely followed Ancona and Caldwell’s (1992) development of boundary-spanning items and selected 13 items from their research to assess the extent to which team members engaged in various boundary-spanning behaviors aimed at establishing and maintaining relationships with key external parties. The wording of some items was adjusted to reflect the particular organization in which we collected data. We further used their factor-analytic results in order to aggregate items into three a priori indicator variables based on their boundary-spanning character: ambassadorial, scouting, and task coordination. The ambassadorial group included five items that reflected both buffering and representational activities. The four items composing the scouting group involved scanning outside the team for ideas about the competition, market, or relevant technology. Finally, the task coordination group included four items that involved coordinating technical or design issues with outside groups. Items were averaged to form the three indicator variables.

Job satisfaction. Survey items for this measure were from the five-item Job Satisfaction Relative to Expectations instrument (Bacharach et al., 1991). The measure assesses the extent of agreement between broad satisfaction elements of a job relative to employee expectation. Previous results showed negative correlations between scores derived from this measure and those from role conflict, role overload, and work–home conflicts (Bacharach et al., 1991).

A measure comprising a subset of the items also showed strong negative correlations with role ambiguity and intent to leave but positive correlations with organizational commitment (Conley & You, 2009).

Demographic variables. We included gender, age, and ethnicity/race as variables in all substantive equations. It is possible that females and non-White team members within a relatively male- and White-dominated organization and/or younger employees in the organization would have comparatively less access to resources that support boundary spanning than do male, White, or more senior employees. There is strong theory and extant evidence to suggest that members of subordinated social identity groups (e.g., women and non-Whites) have less access to resources at work and experience greater challenges and lower job satisfaction, whereas dominant groups (and their members) possess a “disproportionately large share of positive social value” (Amis, Mair, & Munir, 2020), such as power and status. Age can also afford members with greater power and rewards, and a positive correlation between age and job satisfaction has been found in numerous studies in the organizational behavior literature (see Lee & Wilbur, 1985; Rhodes, 1983). We measured age in years; we coded gender using male = 1 and female = 2. Finally, we asked respondents, “What is your ethnicity/race?” and included seven response options: African American/Black, Asian, Caucasian/White, Hispanic/Latino, Native American, Pacific Islander, and Other. Given the preponderance of Caucasian/White respondents, we ultimately coded the measure using Caucasian/White = 1 and all others = 2.

Preliminary Analyses

Table 1 includes means, standard deviations, variable correlations, and reliabilities based on the matched sample of usable data (see Table 1).

We evaluated the quality of our measures by first specifying and assessing a latent variable baseline measurement model using maximum-likelihood estimation methods in Bentler’s (2007) EQS 6.1 program. Construct items and factor groupings were used as indicators for the latent variables (see Figure 1). Overall model fit was evaluated using χ^2/df ratio, the comparative fit index (CFI), and the non-normed fit index (NNFI). In addition, the standardized root mean square residual (SRMR) as well as the root mean square error of approximation (RMSEA) was used to assess model lack of fit. Good model fit can be inferred when the χ^2/df ratio falls below 3.00, CFI and NNFI values exceed .90, and SRMR and RMSEA values are under .10 (Kline, 2005). Measurement parameter estimates were evaluated for significance, and nested-model comparisons were examined using the sequential chi-square difference test (we used an alpha of .05 throughout).

Measurement analyses required complete data at the item level, resulting in a sample size of 164. The measurement model provided good fit to the data, $\chi^2(129, N = 164) = 210.97$, $p < .001$, $\chi^2/df = 1.63$, CFI = .97, NNFI = .96, SRMR = .04, RMSEA = .06, and all standardized factor loadings were significant and ranged from .71 to .96 ($M = .80$). To examine the construct independence of our measures, we compared the baseline measurement model with a series of alternative nested models that constrained theoretically independent constructs to be equivalent (see Table 2). First, a single-factor measurement model specifying perfect correlation among all model latent variables was specified to assess overall

Table 1
Variable Means, Standard Deviations, Correlations, and Reliabilities

Variable	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7
1. Age	41.92	11.11	—						
2. Gender	1.10	0.30	-.12	—					
3. Ethnicity	1.35	0.48	-.12	.00	—				
4. Supportive coaching	3.72	0.69	-.08	-.01	.05	(.88)			
5. Boundary-spanning self-efficacy	3.63	0.82	.12	-.02	-.03	.32**	(.95)		
6. Boundary-spanning behavior	2.62	0.83	-.03	-.13	.18*	.19*	.35**	(.94)	
7. Job satisfaction	3.27	1.08	.01	-.05	.09	.24**	.29**	.29**	(.88)

Note: *N* = 170. Scale reliabilities are along the diagonal in parentheses. Listwise deletion following variable creation.

**p* < .05

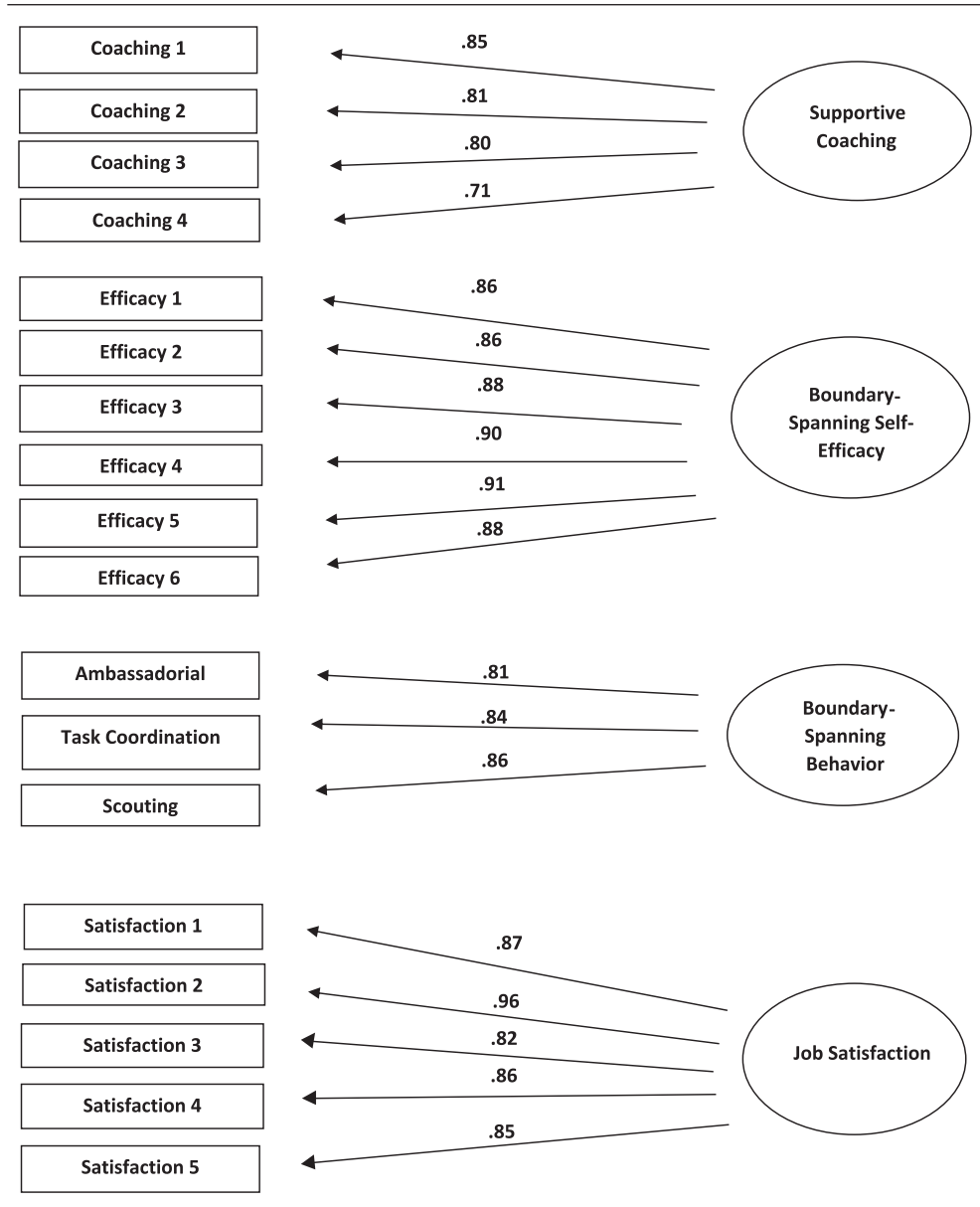
***p* < .01

discriminability. Next, separate tests were conducted to evaluate the conceptual distinction between each of the model constructs. These tests consisted of comparing the baseline model with models that constrained particular constructs to be equal. The single-factor model used to assess overall discriminability (Model 2 in Table 2) poorly accounted for the sample data. These results support the multidimensionality of the model being tested. Subsequent sequential chi-square difference tests showed that constraining combinations of latent variables to be equivalent (see Models 3 to 8 in Table 2) produced significantly worse-fitting models when compared with the baseline measurement model. Moreover, decreases in model CFI indicated a material reduction in model fit for each of the constrained models. These findings provide initial support for the quality of our measures.

Additional evidence pertaining to the convergent and discriminant validity of our constructs is provided in Table 3. All constructs exceed the .70 recommended composite construct reliability threshold as well as the .50 recommended average variance extracted (AVE) threshold (Fornell & Larcker, 1981; Hair, Black, Babin, & Anderson, 2010), further supporting the convergent validity of our measures. Moreover, AVE values exceed the maximum shared variance and average shared squared variance in all of our model constructs, thereby demonstrating their discriminant validity (Hair et al., 2010). Together, these results attest to the psychometric quality of our measures.

We were also concerned about potential method variance effects in our data. We engaged in both pre- and post-data collection actions in order to address possible common-method bias. Following recommendations by Podsakoff and his colleagues (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003; Podsakoff, MacKenzie, & Podsakoff, 2012), we first sought to limit potential method bias by following three specific pre-data collection strategies. First, we made certain to “eliminate common scale properties” (Podsakoff et al., 2012: 550) and created measures with distinct scale endpoints. Two of our constructs, supportive coaching and boundary-spanning self-efficacy, used traditional Likert scales (1 = *strongly disagree* to 5 = *strongly agree*), whereas our other two constructs utilized alternative anchor points (boundary spanning behavior, 1 = *a very small extent* to 5 = *a very great extent*; job satisfaction, 1 = *very dissatisfied* to 5 = *very satisfied*). Next, we worked to ensure the “temporal, proximal, or psychological separation between predictor and criterion” measures (Podsakoff et al.,

Figure 1
Measurement Model Results



2012: 549). As such, we arranged for proximal separation of our measures by including them in distinct surveys and separating them as much as possible within each survey. Finally, we made sure to collect our ultimate criterion measure separately from other measures. This

Table 2
Indices for Nested Sequence of Measurement Models

Model	χ^2	<i>df</i>	CFI	NNFI	SRMR	RMSEA	$\Delta\chi^2$
1. Baseline	210.97	129	.97	.96	.04	.06	—
2. Single-factor model	1406.19	135	.47	.40	.21	.24	—
3. Equate supportive coaching with efficacy	495.20	132	.84	.82	.11	.13	—
Difference between Models 1 and 3	—	—	—	—	—	—	284.23*
4. Equate supportive coaching with boundary-spanning behavior	468.12	132	.86	.83	.12	.12	—
Difference between Models 1 and 4	—	—	—	—	—	—	257.15*
5. Equate supportive coaching with satisfaction	527.06	132	.83	.81	.13	.14	—
Difference between Models 1 and 5	—	—	—	—	—	—	316.09*
6. Equate efficacy with boundary-spanning behavior	426.29	132	.87	.85	.10	.12	—
Difference between Models 1 and 6	—	—	—	—	—	—	215.32*
7. Equate efficacy with satisfaction	913.3	132	.67	.62	.17	.19	—
Difference between Models 1 and 7	—	—	—	—	—	—	702.33*
8. Equate boundary-spanning behavior with satisfaction	448.2	132	.86	.84	.11	.12	—
Difference between Models 1 and 8	—	—	—	—	—	—	237.23*

Note: $N = 164$. CFI = comparative fit index; NNFI = non-normed fit index; SRMR = standardized root mean square residual; RMSEA = root mean square error of approximation.

* $p < .001$

measure, job satisfaction, was included in the second survey, which was administered approximately 3 months and 1 week after the first survey that contained two of the model predictor measures.

Following data collection, we examined results from three separate analyses in order to assess potential method effects. First, we used exploratory factor analysis to determine if Harman's single-factor test would indicate a common-method factor. Results indicated that a single factor explained only 38% of the variance, much less than the 50% threshold established for the test to indicate a method factor (Podsakoff et al., 2003). Next, we compared the baseline measurement model with a model in which the correlations between model factors were specified as equal to unity (i.e., a method factor). As can be seen in Table 2, results from this constrained method factor model, $\chi^2(135, N = 164) = 1406.19, p < .001$, CFI = .47, NNFI = .40, SRMR = .21, RMSEA = .24, along with a subsequent sequential chi-square difference test suggested the method factor model was a significantly worse fit relative to the original model, $\Delta\chi^2(6, N = 164) = 1195.22, p < .001$.

Finally, we also assessed potential method effects by specifying an additional method factor model that included an unmeasured latent-methods factor within our baseline measurement model. We compared our baseline multifactor trait model to the method factor model (multifactor trait plus method-effects model). This specification is recommended to assess the existence and extent of methods effects (e.g., Podsakoff et al., 2003; Williams, Cote, & Buckley, 1989). Importantly, none of the trait measures became insignificant following inclusion of method factor effects. Moreover, the amount of variance accounted for by method

Table 3
Latent Variable Psychometric Quality and Correlations

Variable	Intraconstruct Reliability				Correlations			
	CR	CR	AVE	MSV	1	2	3	4
1. Supportive coaching	.87	.87	.63	.12	—	—	—	—
2. Boundary-spanning self-efficacy	.95	.95	.78	.14	.36*	—	—	—
3. Job satisfaction	.94	.94	.76	.12	.24*	.32*	—	—
4. Boundary-spanning behavior	.87	.87	.70	.14	.24*	.39*	.35*	—

Note: $N = 164$. CR = construct reliability; AVE = average variance explained; MSV = maximum shared squared variance; ASV = average squared variance.

* $p < .05$

effects was very small—it averaged only 7%. This variance accounted for is less than half the amount of method variance in typical behavioral research (Williams et al., 1989) and compares favorably with method factor variance (11%) found in organizational research (Carlson & Kacmar, 2000). Overall, we believe method variance was not a material problem in this study based on results from these post-data collection analyses.

Results

To examine substantive relationships, we considered our data using multilevel latent variable structural equation modeling procedures included in Mplus (Muthén & Muthén, 1998-2017). Two-hundred and fifty-six usable observations were included in these analyses. We used the TYPE=COMPLEX analysis in order to take into account the nonindependence of observations due to cluster sampling. Because respondents were organized into work groups, which may have contributed to systematic bias in our results, we added the effect of the group-level work unit into our model.² The model was assessed based on maximum likelihood estimation using 5,000 bootstrap draws and bias-corrected 95% confidence intervals (CIs). Indirect effects were assessed by examining each CI; any interval that ranges from negative on the low side to positive on the high side indicates an insignificant indirect effect.

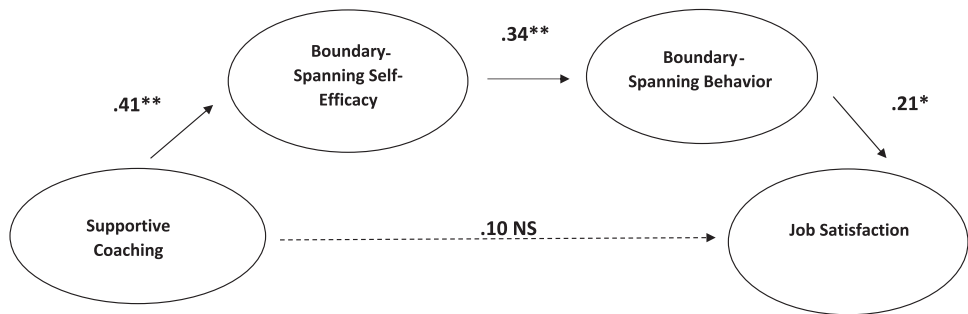
Omnibus model results yielded a good fit to the data, $\chi^2(180, N = 256) = 285.47, p < .001, \chi^2/df = 1.58, CFI = .96, SRMR = .05, RMSEA = .04$. Furthermore, individual model path estimates depicted in Table 4 and Figure 2 indicate strong support for model hypotheses. Specifically, results support Hypothesis 1, in that supportive coaching was positively related to boundary-spanning efficacy ($\beta = .41, SE = .07, p = .000$). Results further demonstrate support for Hypothesis 2 and the direct effect of boundary-spanning self-efficacy on boundary-spanning behavior ($\beta = .34, SE = .08, p = .000$). Results also indicate that boundary-spanning behavior was positively related to job satisfaction ($\beta = .21, SE = .10, p = .039$) providing support for Hypothesis 3. Importantly, the direct effect of supportive coaching on job satisfaction was insignificant ($\beta = .10, SE = .10, p = .351$). Overall, the proposed model predictors explained 15% of the variance in job satisfaction ($R^2 = .15, SE = .07, p = .019$).

Table 4
Path Coefficients and Indirect Effects

Variable	Path Coefficients			Indirect Effects	
	To JS	To BSEFF	To BSB	Estimate	Bias-Corrected Bootstrap 95% Confidence Interval
From					
SC	.10 (.10)	.41 (.07)	.10 (.09)	—	
BSEFF	.20 (.11)	-	.34 (.08)	—	
BSB	.21 (.10)	—	—	—	
Total indirect effect	—	—	—	.13 (.05)	[.05, .21]
Indirect effects					
SC → BSEFF → JSat	—	—	—	.08 (.05)	[.01, .16]
SC → BSB → JSat	—	—	—	.02 (.02)	[-.01, .06]
SC → BSEFF → BSB → JSat	—	—	—	.03 (.02)	[.00, .06]
Direct effect					
SC → JSat	—	—	—	.10 (.10)	[-.07, .27]
Total effect					
SC → JSat	—	—	—	.23 (.09)	[.08, .36]

Note: $N = 256$. Standard errors shown in parentheses. SC = supportive coaching; BSEFF = boundary-spanning self-efficacy; BSB = boundary-spanning behavior; JSat = job satisfaction.

Figure 2
Mediation Model Results



Model $R^2 = .15$ ($p = .019$), CFI = .96, RMSEA = .04

Note: $N = 256$. * $p = .039$; ** $p < .001$.

Table 4 also includes results related to indirect model pathways and Hypothesis 4. Hypothesis 4 stated that supportive coaching would indirectly effect job satisfaction through boundary-spanning self-efficacy and boundary-spanning behavior. As shown in Table 4, supportive coaching was positively associated with boundary-spanning self-efficacy, which was related to boundary-spanning behavior, which was subsequently related to job satisfaction. The indirect effect of supportive coaching on job satisfaction through boundary-spanning self-efficacy and

boundary-spanning behavior was significant ($\beta = .03$, 95% CI [.00, .06]), thereby supporting serial mediation. Moreover, the insignificant direct effect of supportive coaching on job satisfaction ($\beta = .10$, 95% CI [−.07, .27]) indicates that including the two mediators in the model eliminates the direct effect of supportive coaching on job satisfaction. These findings demonstrate support for Hypothesis 4 and the proposed indirect effects specified in the model.

Though not hypothesized here, results in Table 4 also can be used to examine simple indirect pathways included in the model. In particular, the pathway from supportive coaching to job satisfaction via boundary-spanning self-efficacy yielded a significant indirect effect ($\beta = .08$, 95% CI [.01, .16]). However, the pathway from supportive coaching to job satisfaction via boundary-spanning behavior yielded an insignificant indirect effect ($\beta = .02$, 95% CI [−.01, .06]). Together with the results pertaining to model hypotheses, these results attest to the strong influence of efficacy perceptions on subsequent outcomes.

Discussion

The outcome of the indirect-effects model tested here indicates that team leader supportive coaching of boundary spanning positively relates to boundary-spanning self-efficacy among team members. This, in turn, relates to greater team members' engagement in boundary-spanning behaviors, which positively relates to their job satisfaction. Through their boundary-spanning behavior, team members initiate and manage relationships with the team's clients and external suppliers, coordinate efforts with mutually interdependent work teams also inside the organization, and seek out feedback and resources from external subject-matter experts so that work tasks both within and across multiple teams can be accomplished effectively and efficiently (Ancona & Caldwell, 1992). While prior research to date has examined a range of beneficial outcomes associated with boundary spanning for teams and organizations, the calls for greater attention to the drivers of these important behaviors (e.g., de Vries, Walter, Van der Vegt, & Essens, 2014; Joshi et al., 2009; Marrone, 2010) are important to answer. Engagement in these behaviors is challenging for individuals, and some evidence suggests that individual motivation to carry out boundary spanning for one's team and/or organization varies (see Marrone et al., 2007) and may be low for at least some team members (e.g., Lanaj & Hollenbeck, 2015). Traditionally, boundary spanning was viewed as the responsibility and authority of organizational leadership (e.g., Aldrich & Herker, 1977; Katz & Kahn, 1978), and scholars have made notable strides toward understanding the actions and decisions taken by leaders regarding their own boundary-spanning roles and behaviors (e.g., Druskat & Wheeler, 2004; Ernst & Chrobot-Mason, 2011).

Yet this prior research is missing a critical element present in modern organizational designs: The responsibility for boundary spanning in many areas of the organization is increasingly moving to team members and first-line employees (Ancona & Bresman, 2007; Cross et al., 2000). A new and timely theoretical question has thus emerged: How can team leaders influence and motivate team members to carry out important boundary-spanning tasks for their teams? Our study findings suggest a compelling two-part answer. First, targeted supportive-coaching behaviors by team leaders relate to team members' boundary-spanning capabilities independently from the team leader. Second, when more capable (i.e., more efficacious) team members engage in greater boundary-spanning behaviors as a result of their leaders' supportive-coaching behaviors and motivational influence, team members

also increase their job satisfaction. Our findings have theoretical extensions and implications for teams coaching and boundary-spanning research.

Theoretical Implications

First, the teams coaching literature in the organizational behavior domain has largely ignored the impact of supportive-coaching behaviors on team member interactions between or across work teams. Instead, extant studies of a team leader's supportive-coaching behaviors have overwhelmingly focused on factors internal to the team's boundary, such as team member shared leadership (Carson et al., 2007), voice within the team (Farh & Chen, 2018), team commitment (Weer et al., 2016), and team innovation (Rousseau et al., 2013). This internally oriented focus is significant and remains despite that team external relationships are essential to the effectiveness of today's complex and highly interdependent organizations (see Ancona, Backman, & Isaacs, 2019). This study thus extends the focus to the relationship between leader supportive-coaching behaviors and team member efforts to engage effectively across team boundaries.

Second, by clarifying and validating the effect of team leader supportive-coaching behaviors on team member boundary-spanning capabilities, we offer much needed theoretical clarity around the effective leadership of externally dependent work teams. The extant boundary-spanning research examining a team leader's influence on team member boundary-spanning behavior was reviewed a decade ago by Marrone (2010). She found evidence for the influence of directive leadership approaches on driving team member boundary-spanning actions in early foundational work by Ancona (1990) and Ancona and Caldwell (1992) and limited, mixed findings relating to supportive team leadership approaches in later work. Three studies emerged; two found that supportive-coaching behaviors positively influenced teams to actively coordinate or communicate with other work groups (Edmondson, 2003; Tesluk & Mathieu, 1999), while the other found insignificant or negative effects of supportive-coaching behaviors on cross-boundary actions and building relationships with other teams (Druskat & Kayes, 1999). These studies relied on generic assessments of supportive-coaching behaviors, however, rather than a boundary spanning-specific assessment, as we have done here. The specificity of our four-item scale measuring supportive coaching of boundary spanning is an important contribution to future research, as more global/generalized assessments may not have the ability to detect finer-grained theoretical mechanisms underlying the boundary-spanning phenomenon.

Third, our findings that supportive-coaching behaviors relate to team member boundary-spanning behavior through an essential motivational mechanism—increased team member beliefs in their boundary-spanning capabilities (i.e., boundary-spanning self-efficacy; Marrone et al., 2007)—integrates the broader managerial coaching literature and the boundary-spanning literature at a level of detail that has previously been largely neglected in both literatures. Specifying and testing the underlying motivational mechanism answers recent calls for greater theoretical understanding of both (a) the antecedents to team member boundary-spanning behavior as noted earlier (e.g., Joshi et al., 2009; Marrone, 2010) and (b) the specific psychological mechanisms through which ongoing and managerial coaching behaviors lead to valued outcomes (e.g., Dahling et al., 2016). Our findings support the notion that supportive coaching can be a potent source of verbal persuasion regarding team member

capabilities in boundary spanning that motivates them to engage in greater levels of these important activities for their teams and perhaps persist longer in challenging boundary-spanning tasks. These findings complement other work highlighting the role of self-efficacy for explaining the motivational influence of supportive (facilitation-based) coaching in particular, as opposed to more active (guidance) coaching (see Hui & Sue-Chan, 2018), and may help to advance a more unified theory of the mechanisms through which supportive-coaching behaviors relate to productive team member behaviors and positive job attitudes.

Additionally, the question of whether boundary spanning creates role stress at the individual level for team members has been an issue in prior boundary-spanning literature (e.g., Marrone et al., 2007; Singh et al., 1994). The results of this study suggest that as team leaders send strong signals through supportive-coaching behaviors that boundary spanning is valued and worthwhile and give latitude to team members to engage in boundary-spanning tasks as they deem fit, team members may also see an increase in positive attitudinal outcomes at the individual level (i.e., higher job satisfaction), even as they exhibit higher levels of boundary-spanning behavior. Notably, we collected data on boundary-spanning behavior and job satisfaction on the same (i.e., second) survey; while we have good theoretical reason to believe that higher levels of boundary-spanning behavior occur when team members have higher levels of boundary-spanning self-efficacy, as we hypothesized, we cannot rule out the possibility that higher job satisfaction may also lead to higher levels of boundary-spanning behavior. In line with other models of job satisfaction (e.g., Locke, 1970), however, we view the concept of satisfaction as a function of performance leading to valued outcomes that are satisfying to individuals, suggesting that boundary-spanning behavior drives job satisfaction. Our findings provide an additional window into the potentially compelling influence of supportive team coaching behaviors on skill-based and affective outcomes of team members.

Finally, a growing number of studies have delineated facilitative (supportive) from directive (active) coaching (in the latter, leaders directly intervene in team members' taskwork and problem solving) and suggest the distinction is essential yet often overlooked in studies of coaching (e.g., Carson et al., 2007; Hui & Sue-Chan, 2018; Ibarra & Scoular, 2019; Morgeson, 2005; Wageman, 2001). We drew upon this work when conceptualizing and assessing supportive coaching as a set of behaviors that reinforce team members' capabilities to carry out boundary spanning. We further specified that the combination of encouragement and responsibility is vital as it both guides members toward engagement in productive boundary-spanning tasks and empowers members with the authority and control to carry out these actions independently from their leader. Our attention on supportive-coaching behaviors per se and not on active coaching may help to answer calls in the managerial coaching literature to clarify varying coaching dimensions or types and achieve greater consensus as to what behaviors constitute coaching (see Dahling et al., 2016; Ellinger, 2013).

Practical Implications

One implication of our findings is that they offer organizational leaders a deeper and broader understanding of the benefits of team leader supportive coaching behaviors. In business environments increasingly characterized by rapid change and disruption, managers are expected to adopt "coachlike" approaches to their employees that encourage innovation, build commitment, and signal investment in human resources (e.g., Ibarra & Scoular, 2019).

New organizational designs rely on the competencies and self-directed behaviors of the organization's actors, as opposed to hierarchical control, to bring about the cross-boundary coordination and communication activity needed for organizational success (Fjeldstad, Snow, Miles, & Lettl, 2012). That is, meeting organizational objectives increasingly requires that those closest to the work take initiative to promote their team's progress to upper management, secure and exchange needed resources from other units, anticipate environmental changes, and coordinate interdependencies with other work groups. This study indicates that team leaders can employ supportive-coaching behaviors to respond to these organizational needs by reinforcing the capabilities of employees at the lowest levels in the organization to self-organize and to accumulate and share resources directly with multiple external parties who reside across group and organizational boundaries. Moreover, leaders should recognize the primary motivational role of boundary-spanning self-efficacy and apply our findings to tailor their leadership approach and its effect on team member motivation.

Our research also implies that organizations should allow team leaders sufficient latitude to exhibit supportive-coaching behaviors. Encouraging and empowering team members to carry out boundary-spanning tasks independently from the team leader may be challenging for team leaders who have been enculturated to believe that more directive leadership approaches are most effective (see Ancona et al., 2019, for a discussion). If organizations do not work to create a culture and support structure that value empowerment (Ancona et al., 2019; Rapp, Gilson, Mathieu, & Ruddy, 2016), leaders who demonstrate empowerment and encouragement of boundary spanning may be seen by other leaders, and potentially by their own team members, as lacking leadership qualities or competence. This may be particularly true for women team leaders (see Marrone et al., 2018). Our results could help to build organizational cultures valuing supportive coaching by underscoring the importance of these behaviors as a potent dimension of team coaching and clarifying its unique motivational influence on the boundary spanning of team members who are facing tasks that likely require expertise and resources outside their own personal skill sets. Moreover, that team members would exhibit higher levels of boundary-spanning behavior—and feel more satisfied with their jobs in the process—is compelling evidence that organizations should take a strategic interest in developing supportive-coaching capabilities (Ibarra & Scoular, 2019). Equipped with the knowledge of this study's results, team members themselves may be better able to advocate for leaders who exhibit greater supportive-coaching behaviors of boundary spanning for their team or other teams, thus also strengthening the organizational culture's values around supportive coaching.

Limitations and Future Research

There are limitations of the present study. First, while our model proposes serial mediation, we collected data at just two points in time. As we noted earlier, we did so in order to manage the organization's concerns about survey length and mitigate common-method bias (especially important when collecting data examining leader behaviors and outcome variables, as noted by Podsakoff et al., 2012). We did not collect data on the four substantive variables in our study at multiple points in time (i.e., our research was not longitudinal); thus we cannot make causal inferences relative to our results. Moreover, the research design of this study was not experimental or quasiexperimental; as a result, the causal nature that is

suggested in our theoretical model cannot be validated with the existing data set. Future research that is longitudinal and experimental and/or quasiexperimental in a lab or field setting with the same concepts collected at multiple points in time so as to track changes after establishing supportive coaching for boundary spanning is necessary in order to provide further support for the causality of the results described here. In addition, future research, such as a panel design, might be able to capture variable relationships (i.e., cross-lagged and autoregressive effects) that we are not able to assess in the current study.

Second, our use of same-source data collection, despite the fact that we collected data at two points in time, may have led to method bias in our results. Our results across three separate tests, however, did not indicate that method bias was a material problem in this study. Future research should consider incorporating behavioral measures of boundary spanning (e.g., the number of meetings on a given employee's calendar with outside departments or parties) or team member ratings of their peers' boundary-spanning behaviors as another way to avoid common-method bias.

Third, there are current design limitations specifically around the consideration of boundary-spanning self-efficacy. In particular, Lord, Diefendorff, Schmidt, and Hall (2010) called for more empirical work on self-efficacy to be conducted at the within-person level. Future research might consider how supportive coaching helps to develop within-person boundary-spanning self-efficacy, behaviors, and job satisfaction in a dynamic way that unfolds over time. That is, studies might be designed to collect data on the same concepts at multiple time points and use an analytical technique that focuses on the analysis of change over time (e.g., longitudinal growth modeling; Ployhart & Vandenberg, 2010). Furthermore, while our measure of boundary-spanning self-efficacy is consistent with measures utilized in management and applied psychology fields, designed for working professionals who may have limited time to complete a survey with longer instruments (see Maurer & Pierce, 1998), future research should consider additional measurement approaches to further validate our findings. In particular, researchers could consider potential benefits of an alternative boundary-spanning self-efficacy measure aligned with traditional approaches that use a 100-point, 10-unit interval scale and compute composite measures of strength and magnitude (see Bandura, 2006; Lee & Bobko, 1994).

Last, some prior examinations of boundary-spanning behavior in teams have considered the concept at the team level rather than at the individual level (e.g., Marrone et al., 2007, who utilized aggregated team member ratings of boundary spanning). The theoretical emphasis of this research, however, is at the individual level, as we examine how supportive coaching might motivate individual team members to exhibit more boundary-spanning behaviors and how that might influence their individual job satisfaction. Also, unlike the teams studied in Marrone et al. (2007) and other prior research (e.g., Ancona, 1990; Faraj & Yan, 2009), the work units in this study were larger in size and not fully colocated, and some members were part of multiple teams concurrently. It was therefore difficult, if not impossible, for coworkers to observe and assess accurately the extent of each other's boundary spanning or boundary spanning at the team level. Hierarchical linear modeling analysis and our primary Mplus analysis results revealed that our data did not appear to be biased due to between-group variance, suggesting that it was appropriate to test hypotheses at the individual level.³

Moving forward, because our study is among the first to demonstrate the importance of supportive coaching by team leaders regarding a specific task—boundary spanning—our

findings suggest that more theoretical development linking the influence of team leader coaching behaviors to team processes and outcomes could be an important future avenue for research. We see promise in work that explores if certain types of coaching may be more relevant for certain developmental stages of a given team's life span. Additionally, team leader supportive coaching behavior may help teams build collective efficacy, or the team's belief that it can effectively handle certain tasks (e.g., Bandura, 2000; Stajkovic, Lee, & Nyberg, 2009). Transformational leadership, vision, and shared leadership are all positively related to higher levels of collective efficacy (e.g., Fiset & Boies, 2019; Nicolaides et al., 2014; Wang & Howell, 2012); future research could examine whether supportive-coaching behavior influences collective efficacy and whether this could affect other important team processes and outcomes, such as team boundary spanning and performance.

Finally, while it was outside the scope of this article to examine the drivers of respondents' perceptions of team membership boundaries, future studies might explore the role that a supportive team coach of boundary spanning might play in encouraging team members to reflect upon their perceptions of team boundaries and to increase awareness that perceptions may differ from others inside or outside the team. Wimmer, Backmann, and Hoegl (2019), for example, argued members' perceptions of team boundaries are likely to vary in relation to individual-level attention and social categorization processes and that such variation should be welcomed (see also Mortensen & Haas, 2018).

Conclusion

The idea that team leader supportive-coaching behaviors as they relate to boundary spanning specifically may provide a necessary support to help team members navigate increasingly ambiguous and uncertain work-role requirements is a powerful one for organizations. We hope this research has provided additional insight linking supportive-coaching behaviors that encourage boundary-spanning activities, boundary-spanning behaviors themselves, and job satisfaction within an organization that is experiencing the realities of highly technical, complex, cross-functional, and fluid work requirements (e.g., Hackman, 2012; Okhuysen, Lepak, Lee Ashcraft, Labianca, Smith, & Steensma, 2015).

Appendix

Survey Items Used in Study

Company and group area identifiers have been replaced with “###.”

Supportive coaching. Please indicate the extent to which you agree with the following statements in reference to your team's lead. Our lead . . .

1. . . encourages us to engage in an open and honest information exchange with important external parties (such as customers, design suppliers, upper management, external technical experts, and other engineering and/or project teams within and outside of ###).
2. . . promotes frequent collaboration with important external parties.
3. . . grants my team members the autonomy to initiate contact with important external parties when I deem it is necessary to do so.

4. . . . grants my team members the authority to make decisions about when to accommodate external party requests and when to “push back.”

(Responses on this scale ranged from 1 = *strongly disagree* to 5 = *strongly agree*.)

Boundary-spanning self-efficacy. Based on my past experiences working on various project and/or engineering teams thus far at ###, I feel very confident . . .

1. . . . being an advocate for my team to important external parties outside my team (such as clients, suppliers, upper management, external technical experts, and other engineering and/or project teams within and outside of ###) if necessary.
2. . . . soliciting feedback for my team from relevant external parties.
3. . . . representing my team to external parties as we discuss project business.
4. . . . initiating contact with persons outside of the team to discuss team-related problems.
5. . . . maintaining external relationships that might be helpful to my team.
6. . . . establishing connections with those outside the team that can provide the team with guidance/support.

(Responses on this scale ranged from 1 = *strongly disagree* to 5 = *strongly agree*.)

Boundary-spanning behavior. Please indicate to what extent you personally engage in the following behaviors with respect to your team’s project.

Ambassadorial

1. Preventing outsiders from “overloading” the team with too much information or too many requests.
2. “Talking up”/promoting the team to outsiders.
3. Persuading others to support the team’s decisions.
4. Acquiring resources (e.g., access to information, supplies, etc.) for the team.
5. Proactively seeking the advice and support of your team’s manager.

Scouting

6. Reaching out to individuals outside your team that can provide technical ideas/expertise.
7. Scanning the environment inside or outside the organization for technical ideas/expertise.
8. Scanning the environment inside or outside the organization for process ideas/expertise.
9. Finding out what other groups are doing on similar projects.

Task coordination

10. Keeping other groups inside or outside of ### informed of your team’s progress.
11. Negotiating with outside groups for delivery deadlines.
12. Resolving design problems with external groups.
13. Coordinating activities with external groups inside or outside ###.

(Responses on this scale ranged from 1 = *to a very small extent* to 5 = *to a great extent*.)

Job satisfaction. At the present time, how satisfied or dissatisfied are you with . . .

1. . . . the progress you are making towards the goals you set for yourself in your present position at ###?
2. . . . your present job in light of your career expectations?
3. . . . your present job when you compare it to jobs in other organizations?
4. . . . the chance your job gives you to do what you are best at?
5. . . . your present job when you consider the expectations you had when you took the job?

(Responses on this scale ranged from 1 = *very dissatisfied* to 5 = *very satisfied*.)

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Notes

1. Prior to surveying team members, we conducted semistructured interviews with 18 managers and lead engineers in order to better understand the nature of their work and organizational structure and to ensure that previously established measures of boundary-spanning behavior and self-efficacy were appropriate for this organizational context. We learned team members were organized into functional areas similar to large working units or departmental groups (average reported size was 19.1 members). Interview responses also confirmed that boundary-spanning behavior by team members was desired and valued within the organization yet varied between individuals, resulting in organizational management wanting to understand further the sources of such variance. Noted boundary-spanning activities followed earlier boundary-spanning research and included collaborations, negotiations, and transfer of knowledge among multiple and distinct groups of engineering employees, manufacturing employees, and external supplier organizations striving to work collaboratively in designing and manufacturing the lead organization's products.

2. We thank an anonymous reviewer for this suggestion.

3. We used hierarchical linear modeling (Raudenbush & Bryk, 2001) to additionally assess if nonindependence was likely to be problematic for interpreting our results. The null model indicated that no significant variance in our dependent variable, job satisfaction, existed across work units, that is, $\chi^2(79) = 95.04, p = .106$, and one-way ANOVA tests of all substantive variables also yielded nonsignificant results.

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